

RSA Algorithm

1. Aim

To implement the RSA public key cryptographic algorithm for encryption and decryption of a message, and to understand the role of prime numbers, the public key and the private key in asymmetric cryptography.

2. Algorithm

1. Start the program.
2. Read two prime numbers p and q entered by the user.
3. Compute $n = p * q$ and Euler's totient function $\phi = (p - 1) * (q - 1)$.
4. Read the public key e and the private key d entered by the user, chosen such that $(e * d) \bmod \phi = 1$.
5. Read the plaintext message m entered by the user.
6. Encrypt the message using modular exponentiation: $c = m^e \bmod n$.
7. Display the encrypted message.
8. Decrypt the cipher using modular exponentiation: $m = c^d \bmod n$.
9. Display the decrypted message and stop the program.

3. Program

```
#include <stdio.h>

long long power(long long a, long long b, long long mod) {
    long long result = 1;
    while (b > 0) {
        if (b % 2 == 1)
            result = (result * a) % mod;
        a = (a * a) % mod;
        b /= 2;
    }
    return result;
}

int main() {
    long long p, q, n, phi, e, d, m, c;

    printf("Enter Prime Number p: ");
    scanf("%lld", &p);
    printf("Enter Prime Number q: ");
    scanf("%lld", &q);

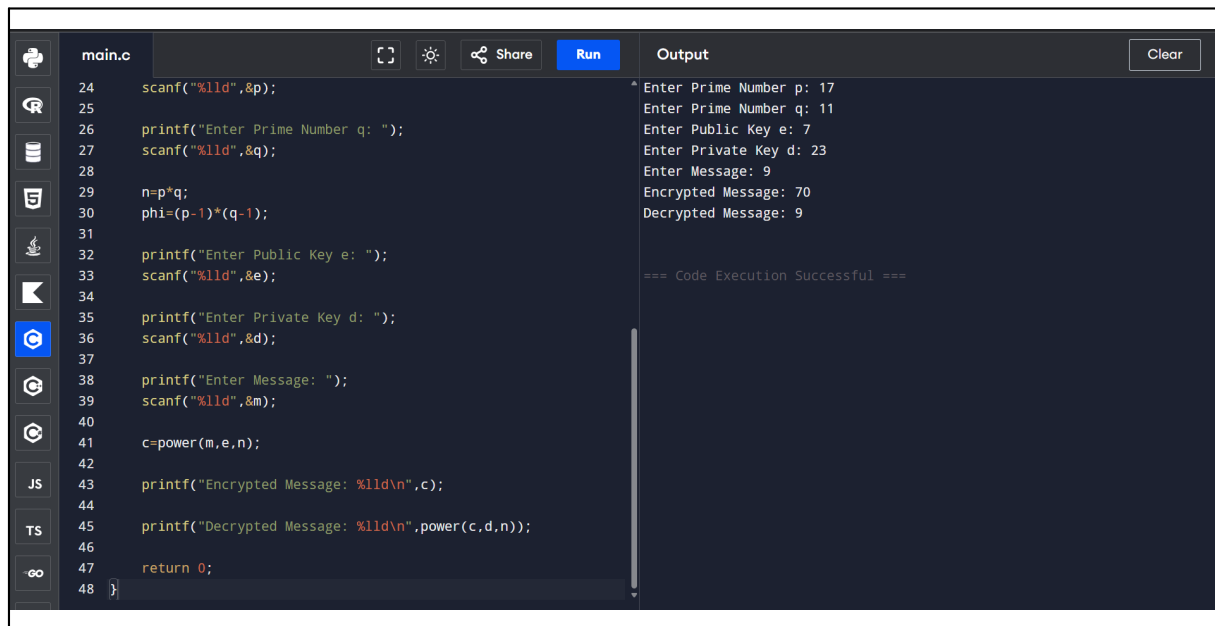
    n = p * q;
    phi = (p - 1) * (q - 1);

    printf("Enter Public Key e: ");
    scanf("%lld", &e);
    printf("Enter Private Key d: ");
    scanf("%lld", &d);
    printf("Enter Message: ");
    scanf("%lld", &m);

    c = power(m, e, n);
    printf("Encrypted Message: %lld\n", c);
    printf("Decrypted Message: %lld\n", power(c, d, n));

    return 0;
}
```

4. Output Screenshot



The screenshot shows a C code editor with a file named 'main.c'. The code implements the RSA algorithm. It prompts the user to enter prime numbers p and q, a public key e, a private key d, and a message m. It then calculates the encrypted message c and the decrypted message. The output window shows the results of the execution.

```
24 scanf("%lld",&p);
25
26 printf("Enter Prime Number q: ");
27 scanf("%lld",&q);
28
29 n=p*q;
30 phi=(p-1)*(q-1);
31
32 printf("Enter Public Key e: ");
33 scanf("%lld",&e);
34
35 printf("Enter Private Key d: ");
36 scanf("%lld",&d);
37
38 printf("Enter Message: ");
39 scanf("%lld",&m);
40
41 c=power(m,e,n);
42
43 printf("Encrypted Message: %lld\n",c);
44
45 printf("Decrypted Message: %lld\n",power(c,d,n));
46
47 return 0;
48 }
```

Output:

```
Enter Prime Number p: 17
Enter Prime Number q: 11
Enter Public Key e: 7
Enter Private Key d: 23
Enter Message: 9
Encrypted Message: 70
Decrypted Message: 9

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter Prime Number p: 7
Enter Prime Number q: 11
Enter Public Key e: 7
Enter Private Key d: 43
Enter Message: 9
Encrypted Message: 37
Decrypted Message: 9
```

6. Result

The RSA public key cryptographic algorithm was implemented and executed successfully, and the original message was correctly recovered after encryption and decryption.